

Introduction to Sensors

Robots in Context

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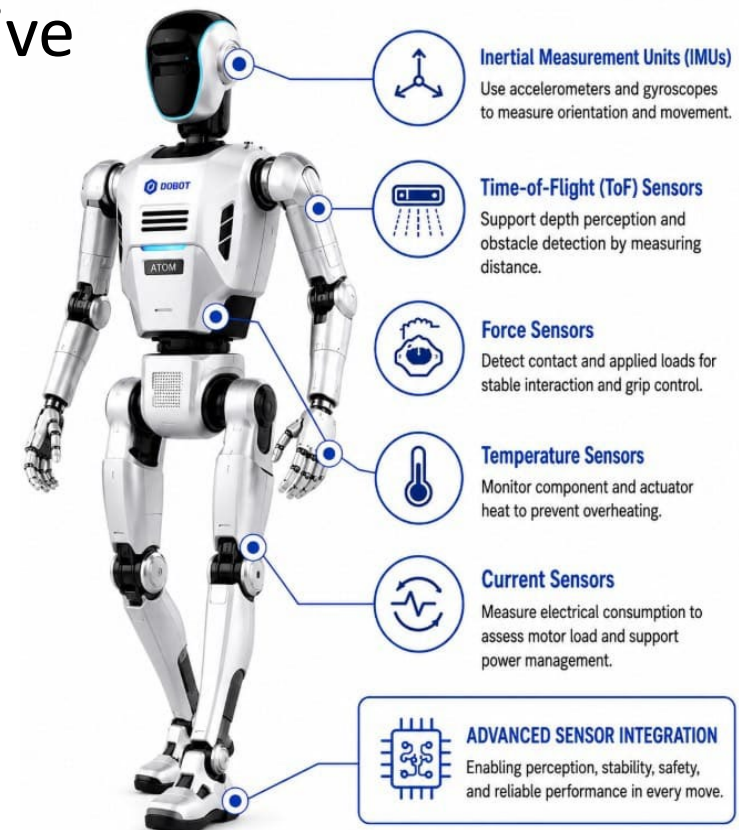
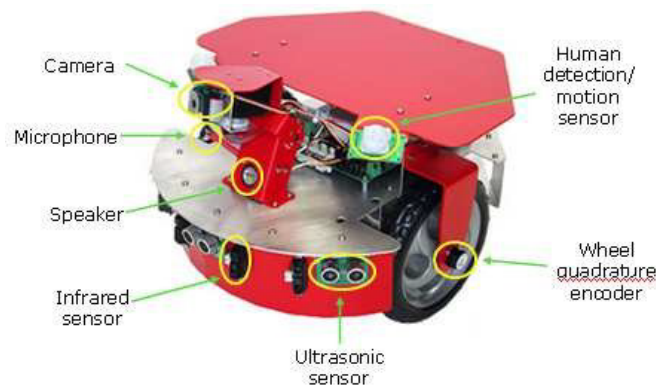
Sensors

- The robots link to the real world
- Sensor classification
 - Modalities – The type of input measured
 - **Proprioceptive vs Exteroceptive**
 - **Passive vs. active**
 - Global vs. Local
 - Error conditions

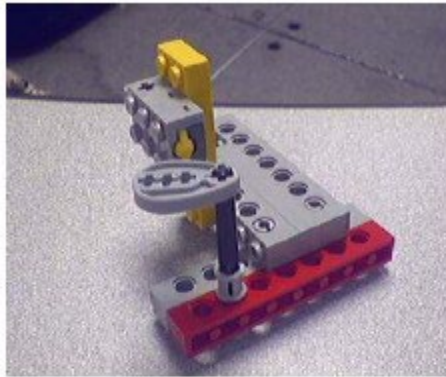


Classify Sensors in NAO

- Modalities – The type of input measured
- Proprioceptive vs Exteroceptive
- Passive vs. active
- Global vs. Local



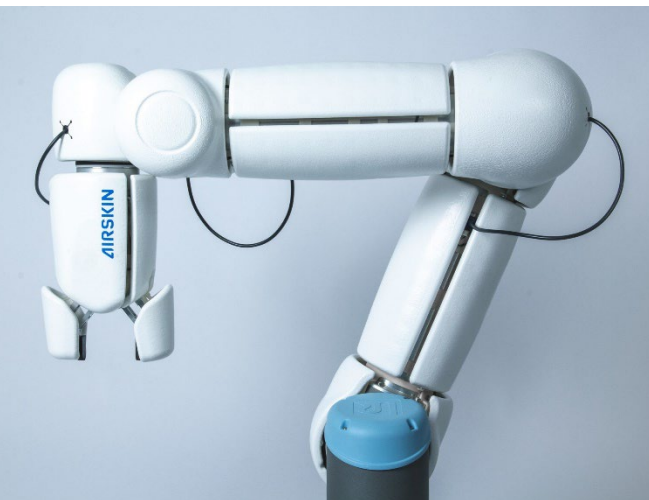
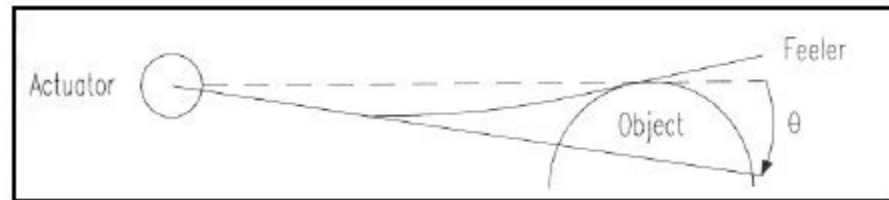
Tactile sensors



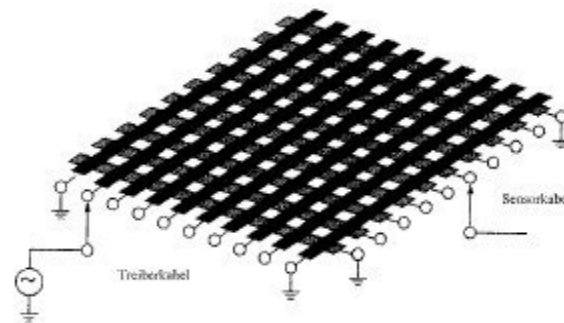
on/off switch

as a low-resolution encoder...

analog input: "Active antenna"



Airskin



Capacitive array sensors



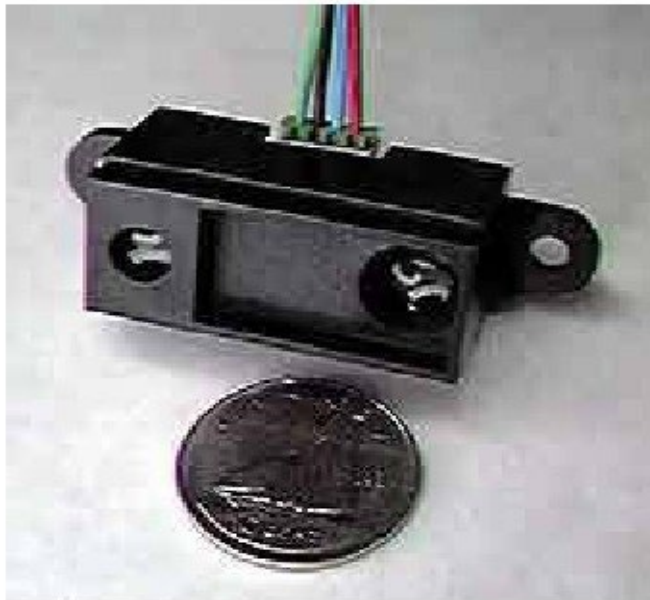
Resistive sensors



Limited/Coarse Range Sensors

IR Sensors and Sonars

“Noncontact bump sensor”

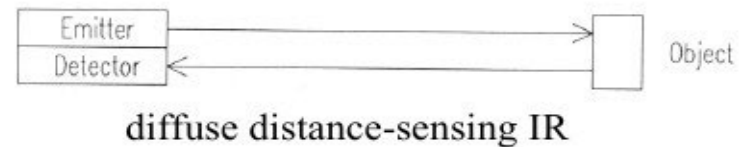
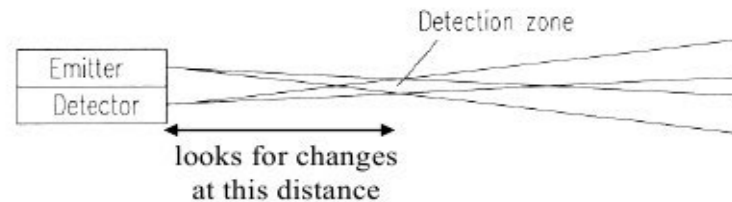


IR emitter/detector pair

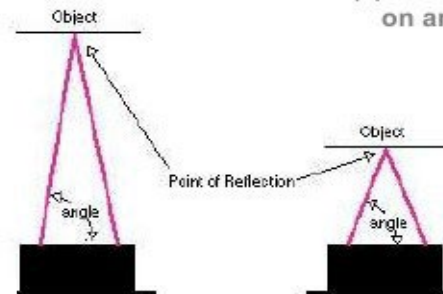


(1) sensing is based on light intensity.

“object-sensing” IR



(2) sensing is based on angle received.



Range Sensors



LIDAR – Laser scanners



Structured Light



Triangulating



Stereo Camera



LIDAR - Principle

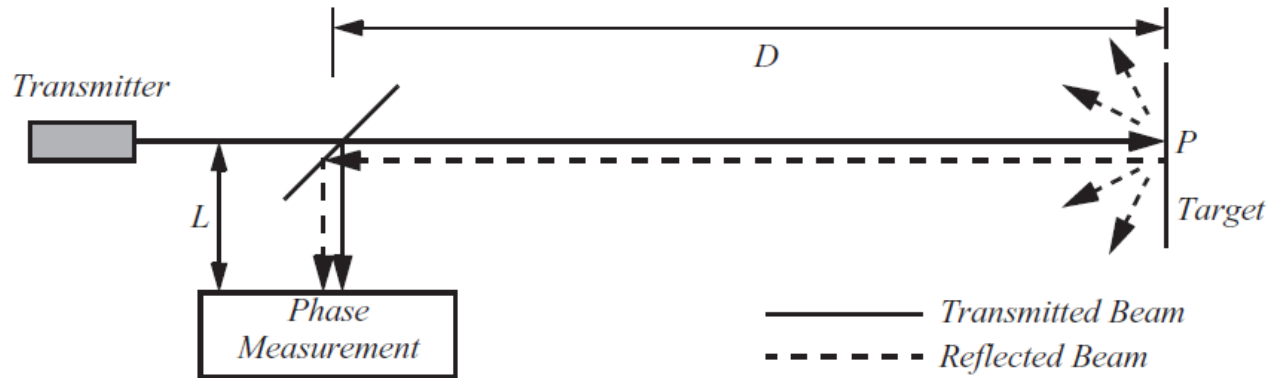


Figure 4.9
Schematic of laser rangefinding by phase-shift measurement.

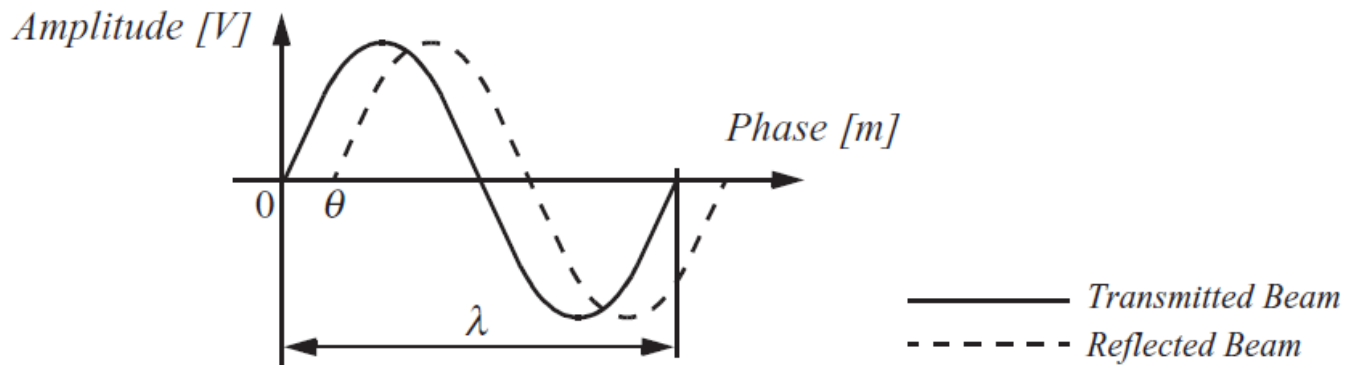
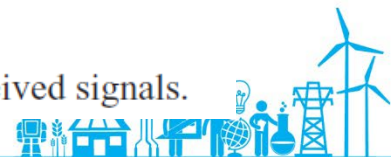
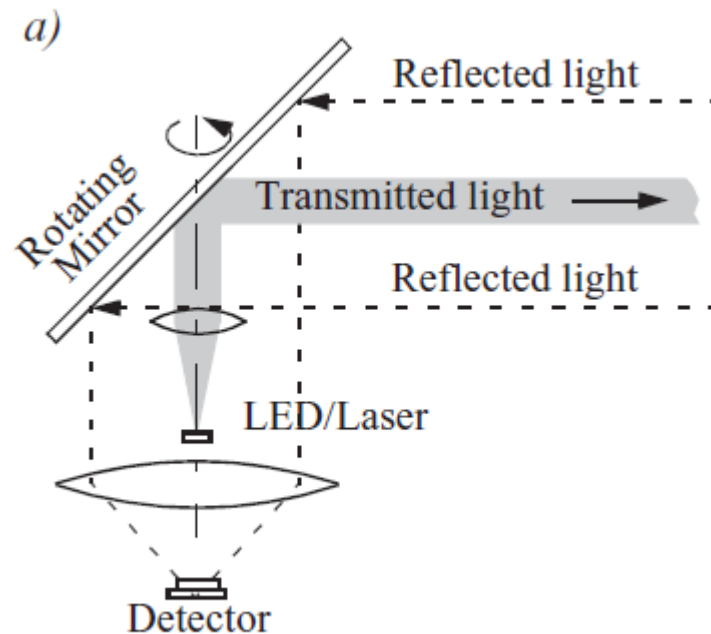


Figure 4.10
Range estimation by measuring the phase shift between transmitted and received signals.



LIDAR - Principle



Triangulating

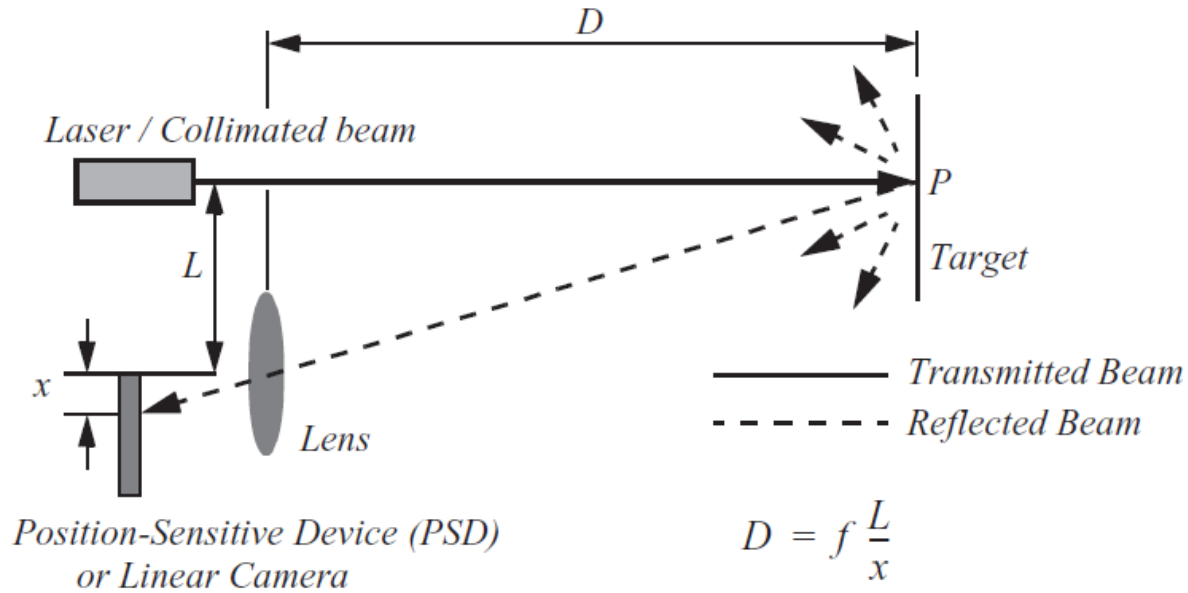


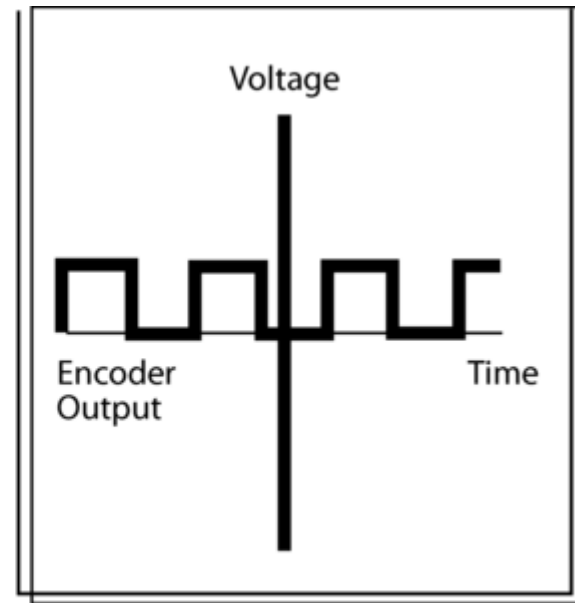
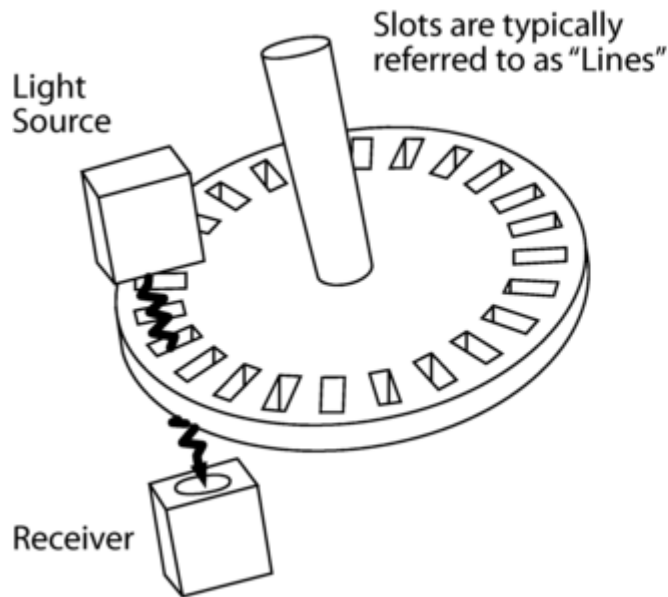
Figure 4.13
Principle of 1D laser triangulation.

Vision Sensors

- Grayscale vs. Color
- Visible light vs. Near infrared
- Lighting
- Filters



Encoders



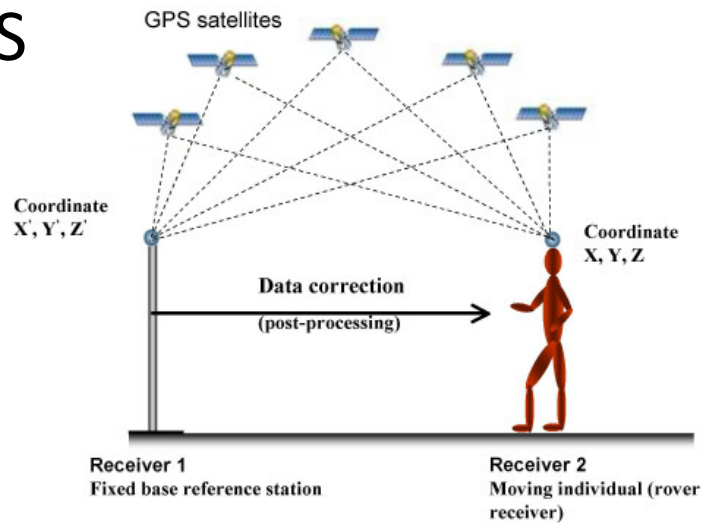
Pulses Indicate Position
Pulses/Sec Indicate Speed

[<http://www.ab.com/en/epub/catalogs/12772/6543185/12041221/12041235/Encoders.html>]



GPS

- Indoor
- Outdoor
 - GPS
 - Differential GPS



Source: https://www.researchgate.net/figure/Differential-GPS-principles-The-satellites-are-viewed-by-both-receivers-located-closed_fig2_7622291



A professional video rig featuring two RED PRO 5.0 cameras mounted side-by-side on a sliding rail. The cameras are equipped with large lenses and external monitors. The rail is supported by a Sachtler base, which includes a sliding mechanism and leveling feet. The entire setup is designed for smooth, controlled camera movement during filming.



Sensor Noise and Errors

Reflection: Consider error sources and noise for the different sensors

- Tactile
- Limited/Coarse range sensors
- Range Sensors
- Vision sensors
- Encoders

